# Presentation

## course "Essentials of computing systems"

Feb.2025

## course

- This course was conceived for programmers and software engineers who want to comprehend how the components of a computer work and how they affect the correctness and performance of their programs.
- Understanding how a computer works is fundamental for software engineers to comprehend the principles governing the execution of the programs they develop/maintain.
- It is also a fundamental knowledge if one wants to optimise the performance of a program, to write a compiler, or to develop an embedded system.
- This course is focused on describing the fundamental aspects of computers, by studying their organisation and structure.

## topics

- Major pedagogical units:
  - Computer systems
  - Representation of information
  - Representation of numbers
  - IA32 instruction-set architecture
  - Cache memory
  - Code optimisations
- We assume that students know how to program a computer in a high-level language (e.g., C).
- Part of the contents of the course will be addressed in Laboratórios de Informática II.
- The contents of this course is continued in "Arquitetura de Computadores" (2.º ano / 1.º semestre).

# professors

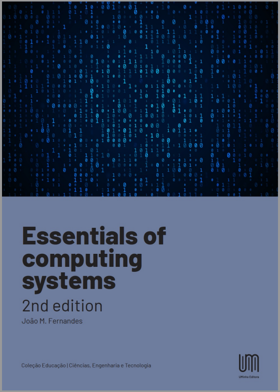
| | | |
|---|---|---|
| João M. Fernandes | JMF | T |
| André L. Ferreira | ALF | PL6 PL9 |
| António J. Esteves | AJE | PL3 PL4 PL8 PL10 |
| Manuel J. Alves | MJA | PL1 PL2 |
| Paulo R. Sousa | PRS | PL5 PL7 |
| André M. Pereira | AMF | TP1 TP2 (LEFIS) |

Professors can be reached through email, but they cannot guarantee immediate answer.

# e-book

- A 100-page e-book was prepared to support the course.
- The e-book is free (published by UMinho-Editora).
- Students must read the book.
- Ambitious students must read other materials (e.g. suggested readings).

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# T-classes

- Slides in English / Speech in Portuguese.
- T-classes serve to expose/discuss course contents.
- Contents are totally described in the e-book.
- Only classes T1 (Mon 10-11; Thu 12-13) are guaranteed to be delivered.
- Classes T2 (Mon 16-17, Tue 12-13) only if T1 classes are full.

# PL-classes

- PL/TP-classes serve to consolidate skills and knowledge.
- Classes will follow the TBL (team-based learning) regimen.
- Participation in 8 TBL PL/TP-classes is mandatory.
- Students are expected to prepare the PL/TP-classes before entering the room.
- Solutions to exercises are provided in the e-book.
- Some classes require access to a computer (`gcc` compiler or access to `godbolt.org`).

PL3 AJE Mon 11-13
PL8 AJE Tue 09-11
PL9 ALF Tue 09-11
PL4 AJE Tue 16-18
PL6 ALF Tue 16-18
PL1 MJA Wed 09-11
PL2 MJA Wed 11-13
PL7 PRS Thu 10-12
PL5 PRS Fri 14-16
PL10 ALF Fri 14-16

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TP1 AMP Tue 14-16

TP2 AMP Tue 16-18

# schedule

| semana | segunda | Teor. | quinta | Teor. | PL | |
|---|---|---|---|---|---|---|
| 1 | 2025-02-03 | Apresent. | 2025-02-06 | Ch1 | | |
| 2 | 2025-02-10 | Ch1 | 2025-02-13 | Ch2 | TBL Ch1 | |
| 3 | 2025-02-17 | Ch2 | 2025-02-20 | Ch3 | TBL Ch2 | |
| 4 | 2025-02-24 | Ch3 | 2025-02-27 | Ch3 | TBL Ch2 | |
| 5 | 2025-03-03 | Ch3 | 2025-03-06 | Ch3 | TBL Ch3 | |
| 6 | 2025-03-10 | Ch4 | 2025-03-13 | Ch4 | TBL Ch3 | |
| 7 | 2025-03-17 | Ch4 | 2025-03-20 | Ch4 | TBL Ch3 | |
| 8 | 2025-03-24 | não há aulas | 2025-03-27 | não há aulas | | |
| 9 | 2025-03-31 | Ch5 | 2025-04-03 | Ch5 | TBL Ch4 | |
| 10 | 2025-04-07 | Ch5 | 2025-04-10 | revisão testes | TBL Ch4 | teste sábado 2025-04-12 |
| 11 | 2025-04-14 | não há aulas | 2025-04-17 | não há aulas | | |
| 12 | 2025-04-21 | não há aulas | 2025-04-24 | Ch6 | TBL Ch5 | feriado sexta 25.04 |
| 13 | 2025-04-28 | Ch6 | 2025-05-01 | não há aulas | TBL Ch5 | feriado quinta 01.05 |
| 14 | 2025-05-05 | Ch6 | 2025-05-08 | folga | TBL Ch6 | |
| 15 | 2025-05-12 | | 2025-05-15 | enterro Gata | | |
| 16 | 2025-05-19 | revisão testes | 2025-05-22 | revisão testes | Debugger | última semana aulas |
| 17 | 2025-05-26 | | 2025-05-29 | | | teste quarta 2025-05-28 |

exame quarta 2025-06-18

## assessment

- Final mark is given by:
  - test #1 (50%)
  - test #2 (50%)
- An overall minimum of 9.50 points is needed to APPROVE.
- Participation in at least 8 of the 11 TBL classes is a mandatory condition to APPROVE.
- Marks are rounded up
  $11.00 \to 11 \quad 11.20 \to 12 \quad 11.50 \to 12 \quad 11.99 \to 12$
- Students that attend both T- and PL/TP-classes and that prepare themselves conveniently should approve.
- (2023/24) <10 52 10-12 113 13-15 76 16-18 16 20 1
  Success rate: 206/258 (80%)